

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION

WINTER SEMESTER, 2023-2024

DURATION: 1 HOUR 30 MINUTES

FULL MARKS: 75

SWE 4301: Object Oriented Concepts II

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) You have a staircase with n steps. You start from the bottom (step 1) and move up to the top (step n). After you reach the top, you descend all the way back to the bottom. You wrote the program (given in Code Snippet 1) that would print each step as you go up and then print each step again as you come down.
Define clean code and explain why the code in Code Snippet 1 would be considered a costly code to write.

2 + 3
(CO1)
(PO1)

```

1 class StairCase{
2     int n;
3     private List<String> steps;
4     StairCase(int st){
5         this.n = st;
6         initializeSteps();
7     }
8     private void initializeSteps() {
9         steps = new ArrayList<>();
10        for (int i = 1; i <= n; i++)
11            steps.add(formatStep("Going up", i));
12        for (int i = 1; i <= n; i++)
13            steps.add(formatStep("Coming down", n - i + 1));
14    }
15    private String formatStep(String direction, int step) {
16        return String.format("%s: Step %d", direction, step);
17    }
18    public void stepCounter() {
19        steps.forEach(System.out::println);
20    }
21 }

```

Code Snippet 1: A StairCase class in Java for Question 1.a

- b) You are asked to implement a data structure called MinStack that supports basic functionality of a stack like `push()`, `pop()`, and extra functionality like `getMin()` for retrieving the minimum element, all in constant time $O(1)$.
Explain how you would follow Test-Driven Development (TDD) with at least three test cases.
- c) "Technical debt is not just bad code." — Do you agree with this statement? Justify your answer with example.

9
(CO2)
(PO1)5
(CO1)
(PO1)

2. a) Consider the classes in Code Snippet 2 to answer the following questions:

```
1 class Character {
2     String name;
3     String characterClass;
4     int magicalPower;
5     int stamina;
6     int level;
7     boolean exh;
8     // Constructor and getters/setters omitted for conciseness
9 }
10 public class RPGCalculator {
11     public Map<String, String> healthStates(List<Character>
12         characters) {
13         return characters.stream()
14             .collect(Collectors.toMap(character -> character.name,
15                 character -> this.healthState(character)));
16     }
17     public Map<String, Double> powerLevels(List<Character> characters) {
18         return characters.stream()
19             .collect(Collectors.toMap(character -> character.name,
20                 character -> this.calculatePowerLevel(character)));
21     }
22     private String healthState(Character character) {
23         switch (character.characterClass) {
24             case "Warrior":
25                 return (character.stamina > 50) ? "healthy" : "
26                     wounded";
27             case "Mage":
28                 return (character.magicalPower > 75) ? "energized" :
29                     "drained";
30             case "Rogue":
31                 return (character.stamina > 40 && !character.exh) ? "
32                     agile" : "fatigued";
33             default:
34                 return "unknown";
35         }
36     }
37     private Double calculatePowerLevel(Character character) {
38         switch (character.characterClass) {
39             case "Warrior":
40                 return character.level * 10.0 + character.stamina;
41             case "Mage":
42                 return character.level * 8.0 + character.magicalPower
43                     * 1.5;
44             case "Rogue":
45                 return character.level * 7.0 + (character.exh ? 0.0 :
46                     character.stamina * 1.2);
47             default:
48                 return null;
49         }
50     }
51 }
```

Code Snippet 2: Character and RPGCalculator classes in Java for Question 2.a

- | | |
|---|----------------------|
| i. Identify the code smells from Code Snippet 2 and mention the line number where a particular code smell is found. | 10
(CO4)
(PO2) |
| ii. Identify the design principles violated in Code Snippet 2 and explain how the code violates them. | 6
(CO4)
(PO2) |
| iii. The clients extended their feature for a new type of character with the following requirements, | 4
(CO3)
(PO1) |
| <ul style="list-style-type: none"> • New character name would be "Goblin", • A goblin's health state should be determined by its stamina: <ul style="list-style-type: none"> – If stamina is greater than 30, the Goblin is "sneaky." – If stamina is 30 or below, the goblin is "weak." • The power for Goblin will be calculated with the expression:
 $\text{power} = \text{level} * 5.0 + \text{stamina} * 0.8 + (\text{exhausted} ? 0.0 : 10.0)$ | |
| Implement the extended feature in Code Snippet 2 (write only the classes/methods requiring modification) and state the number of lines needed to be changed. | |
| iv. You decided that the code in Code Snippet 2 requires refactoring as it is harder to extend the feature. Draw a class diagram or write a code outline with empty methods that reflect the expected structure. Your answer should reflect the client's extended requirement in Question 2.a)iii. | 10
(CO4)
(PO2) |
| b) You are working on a project for a food delivery app. But previous developers were not mindful about writing clean code. The project is now facing a lot of bugs, so you refactored the modules related to the bugs. Explain how you would know whether you have refactored it in a right way. | 7
(CO1)
(PO1) |
| 3. a) Explain the relation between Liskov Substitution Principle and Dependency Inversion Principle, and give a code example where both design principles are violated. | 7
(CO1)
(PO1) |

b) Consider the classes in Code Snippet 3 and Code Snippet 4 to answer the following questions:

```

1 class EmailSender {
2     public void sendEmail(String message) {
3         System.out.println("Sending email: " + message);
4     }
5 }
6 class SMSSender {
7     public void sendSMS(String message) {
8         System.out.println("Sending SMS: " + message);
9     }
10 }
11 class Logger {
12     public void log(String message) {
13         System.out.println("Logging: " + message);
14     }
15 }

```

Code Snippet 3: EmailSender, SMSSender, Logger classes in Java for Question 3.b

```

1 class NotificationService {
2     private EmailSender emailSender;
3     private SMSSender smsSender;
4     private Logger logger;
5     public NotificationService() {
6         this.emailSender = new EmailSender();
7         this.smsSender = new SMSSender();
8         this.logger = new Logger();
9     }
10    public void sendNotification(String message, List<String> channels) {
11        for (String channel : channels) {
12            switch (channel) {
13                case "email":
14                    emailSender.sendEmail(message);
15                    break;
16                case "sms":
17                    smsSender.sendSMS(message);
18                    break;
19                default:
20                    System.out.println("Unsupported channel:" + channel);
21            }
22        }
23        logger.log("Notification sent via " + channels + ":" + message);
24    }
25 }

```

Code Snippet 4: NotificationService class in Java for Question 3.b

- i. Identify how the current implementation of the NotificationService class violates the Dependency Inversion Principle (DIP). 5
(CO4)
(PO2)
- ii. Implement the DIP with dependency injection in Code Snippet 3. You should write your code with clear indication how you have applied DIP and dependency injection. 7
(CO3)
(PO1)